

Personal Project (Mar 2024 – Present)

Job-Seeker (Autonomous AI Career Agent)

- **Link:** Private
- **Timeline:** April 2026 – Present
- **Tech Stack:** Python, FastAPI, LangGraph, LangChain, OpenAI, Claude, Gemini, Multi-Agent Systems, Agentic AI, LLM Orchestration, RAG (Retrieval-Augmented Generation), Prompt Engineering, Structured Outputs, Function Calling, Tool Calling, MCP (Model Context Protocol), A2A (Agent-to-Agent Communication), Vector Databases (Qdrant, Pinecone), Embeddings, Semantic Search, Memory Systems, Long-Term Agent Memory, Workflow Orchestration, PostgreSQL, Redis, Playwright, Gmail API, Google Calendar API, LinkedIn Integration, Naukri Integration, ATS Integrations, React, Next.js, Docker, Kubernetes, CI/CD, Observability & Tracing.
- **About:** Developed an AI-powered autonomous job search platform designed to act as a personal career agent that automates the entire job acquisition lifecycle. Built a multi-agent architecture consisting of specialized agents for **Job Discovery, Resume Generation, ATS Optimization, Application Submission, Recruiter Communication, Interview Coordination, Application Tracking, and Career Workflow Automation**. Implemented a **Multi-LLM orchestration layer** leveraging OpenAI, Claude, and Gemini models with dynamic routing, fallback strategies, structured outputs, and tool-calling capabilities to optimize cost, performance, and reliability.
- Designed advanced **Agentic AI workflows** using LangGraph with stateful execution, memory persistence, agent collaboration, human-in-the-loop approvals, and fault-tolerant recovery mechanisms. Integrated **RAG pipelines, semantic search, vector retrieval, embeddings, and long-term memory systems** to personalize job matching, resume generation, recruiter interactions, and application strategies based on historical user behavior.
- Built intelligent connectors for **LinkedIn, Naukri, Email Platforms, Google Workspace, and external ATS systems** enabling autonomous job discovery, application submission, recruiter engagement, and interview scheduling. Developed AI-powered resume generation and ATS optimization pipelines capable of producing job-specific resumes with tailored skills, achievements, and keyword alignment.
- Integrated **Gmail and Calendar automation** for recruiter communication, automated follow-ups, interview scheduling, reminder creation, meeting coordination, and status tracking. Implemented robust guardrails covering **document sharing policies, privacy controls, compliance validation, approval workflows, hallucination prevention, and sensitive action protection**.
- Created a unified dashboard for monitoring application pipelines, recruiter engagement metrics, interview progress, agent activities, workflow execution states, and overall job search performance with the objective of moving candidates from job discovery to interview scheduling with near-zero manual effort.

Personal Project (Mar 2024 – June 2026)

Local ERP

- **Link::** <https://github.com/vKrypto/marg-erp>
- **Timeline:** Jan 2026 – March 2026
- **Tech Stack:** Node.js, Express.js, React.js, MySQL, HTML, CSS, Bootstrap, Material UI, Browser-SQL
- **About:** Developed a comprehensive healthcare ERP platform and cost-effective alternative to MARG ERP for hospitals, clinics, pharmacies, and healthcare providers. Implemented a hybrid **offline/online architecture** enabling uninterrupted operations and real-time data synchronization. Built a unified platform for **Inventory Management, Billing & Invoicing, Patient Management, Appointment Scheduling, Reporting & Analytics, Supplier Management, Medicine Stock Tracking, Purchase & Sales Management, User & Role-Based Access Control, and Financial Record Management**. Delivered a responsive and user-friendly interface that streamlined healthcare operations, reduced administrative overhead, and improved operational efficiency across medical facilities.

Project Portfolio – Ashutosh Verma

Fluxon (Mar 2024 – June 2026)

FluxMetrics – AI Governance & Cost Intelligence Platform

- **Link:** <https://metrics.fluxon.io/>
- **Tech Stack:** NestJS, TypeScript, PostgreSQL, BigQuery, GCP, Docker, CI/CD, GenAI, RBAC, Observability
- **About:** Architected and built an enterprise-wide AI governance, cost intelligence, and observability platform serving **400+ employees** across **20+ AI platforms**, including Claude, OpenAI, and internal AI services. Aggregated and analyzed **millions of telemetry and token-consumption events** to provide centralized governance, spend optimization, forecasting, anomaly detection, and executive-level AI adoption insights.

Key Highlights

- Built centralized AI cost analytics with **token-level visibility**, team-wise usage breakdowns, budget controls, forecasting, utilization insights, and executive reporting dashboards.
- Developed **GenAI-powered anomaly detection systems** to identify unusual usage spikes, token consumption anomalies, cost outliers, inefficient prompt patterns, and abnormal platform behavior before budget thresholds were breached.
- Implemented intelligent alerting workflows with configurable **hourly, daily, weekly, and monthly monitoring windows**, automated escalations, and AI-generated cost optimization recommendations.
- Designed enterprise-grade **RBAC, audit logging, governance controls, and team-level visibility**, enabling secure access management and organization-wide AI governance.
- Processed and analyzed **millions of AI telemetry events** across multiple providers to deliver actionable utilization and ROI insights.
- Integrated **Claude AI** into engineering workflows for automated pull request reviews, security validation, code quality analysis, documentation feedback, and fix recommendations, reducing manual review effort by **~90%**.
- Enabled leadership teams to monitor AI adoption, platform utilization, spend efficiency, and governance compliance through a unified observability platform.

NAV India (5 Aug 2024 – Feb 2026)

Accounts API Optimization

- **Tech Stack:** Golang, REST APIs, DB Optimization, Profiling, Caching
- **About:** Transformed Accounts API performance from **p90: 2s, p99: 60s** to **p95: 2ms, p99: 40–50ms**. Applied deep profiling, query optimization, and caching strategies to deliver extreme-scale performance.

AI Chatbot & Semantic Search

- **Link:** <https://github.com/vKrypto/semantics-search>
- **Tech Stack:** FastAPI, Elasticsearch, Vector DB, Transformers, OpenAI/Ollama
- **About:** Developed a **retrieval-augmented chatbot** using semantic embeddings and vector search, enabling intelligent contextual Q&A.

Optimize Data Fetch from Exchanges

- **Tech Stack:** FastAPI, Python AsyncIO, Concurrency (10k+ jobs), Caching, Profiling
- **About:** Cut job runtimes from **200s → 5–9s**; scaled system to reliably handle **~10k concurrent jobs** and **~500k API requests per minute** with robust async pipelines.

Python Version Upgrade

- **Tech Stack:** Python 3.x, Dependency Refactor, CI/CD, Test Automation

- **About:** Upgraded legacy Python services; resolved deprecated APIs, migrated dependencies, and hardened CI/CD pipelines with automated unit/integration testing.

Full-stack Observability & Health Monitoring

- **Tech Stack:** Grafana, Prometheus, Kibana, Jaeger, Graylog, Elasticsearch, cAdvisor, Node Exporter, Docker
- **About:** Implemented a **health monitoring agent** across all services and infra components. Integrated system + container exporters with central log aggregation, tracing, and dashboards. Reduced MTTR and improved reliability of distributed workloads.

.NET Legacy Code Refactor (incl. Data Sync Re-engineer)

- **Tech Stack:** .NET, SQL Server (Stored Procedures, ETL), Windows Services, Design Patterns (Repository, Factory, Dependency Injection)
- **About:** Re-architected legacy .NET data synchronization pipelines. Applied design patterns to improve throughput, maintainability, and reliability of critical workflows.

ShyftLabs (1 Nov 2023 – 5 Aug 2024)

FastAPI → TDD Migration

- **Tech Stack:** FastAPI, Python, SQLite (test harness), Pytest, Tox, GitLab CI/CD
- **About:** Converted FastAPI project to strict **TDD**; achieved **98%+ coverage** and uncovered **90+ critical bugs** before production.

CI/CD with Tox and GitLab Pipelines

- **Tech Stack:** Tox, GitLab CI/CD, Docker
- **About:** Automated build, test, and deployment pipelines for consistent, reliable delivery.

Pricing Change Implementation (Petco)

- **Tech Stack:** Python, SQL, API Development
- **About:** Delivered pricing-change engine for **offline US pet stores (Petco)**; integrated APIs with enterprise retail systems.

Data Pipelines for Event Logs

- **Link:** <https://www.trycarter.com/>
- **Tech Stack:** Python, MongoDB, Kestra, Airbyte, Airflow, Batch & Streaming Infra
- **About:** Built fault-tolerant pipelines to process **billions of events daily** using hybrid batch/streaming orchestration frameworks.

ORNAZ (1 June 2018 – 1 Nov 2023)

Desktop E-commerce Replatform (PWA)

- **Timeline:** Apr 2021 – Nov 2023
- **Tech Stack:** Next.js (SSG/SSR/ISR), Node.js, Nginx, CDN
- **About:** Built PWA with **Lighthouse 100/100** and **<0.5s load times**. Integrated multiple payment gateways and improved SEO/UX drastically.

Monolith → Microservices Migration

- **Timeline:** 2021 – 2023
- **Tech Stack:** FastAPI, Django, Docker Swarm, gRPC/REST, PostgreSQL (HA 2M/3S), Celery, RabbitMQ
- **About:** Migrated monolith into **4 core services + workers** with HA database clusters. Achieved **99.99% uptime** with Docker Swarm orchestration.

Observability Platform

- **Timeline:** 2021 – 2023
- **Tech Stack:** Grafana, Prometheus, ELK Stack, Jaeger, Sentry
- **About:** Built centralized monitoring for metrics, logs, and traces. Provided single-pane observability across microservices and APIs.

Diamond Compare Tool

- **Timeline:** 2022 – 2023
- **Tech Stack:** Next.js, FastAPI, PostgreSQL, Elasticsearch, SSR/ISR
- **About:** Created diamond comparison app, increasing engagement **+35%** and conversion **+20%** with real-time indexed search.

Campaign Orchestrator

- **Timeline:** 2022 – 2023
- **Tech Stack:** Node.js, Next.js, Redis, Webhooks, Email/SMS APIs
- **About:** Built trigger-based orchestration engine for campaigns; improved conversions with real-time customer journeys.

Virtual Try-On (Cross-Platform Native)

- **Timeline:** 2020 – 2021
- **Tech Stack:** Java (Android), Swift (iOS), C++ (MediaPipe AR Engine), Web (JS/WebGL + WASM)
- **About:** Built AR try-on apps for **Android, iOS, and Web** with 90% shared C++ AR core. Delivered real-time jewelry trial; boosted engagement **50%** and sales conversion **30%**.

Virtual Trial Room (Web)

- **Timeline:** 2020 – 2021
- **Tech Stack:** JavaScript/WebGL, MediaPipe (C++), WASM, CDN, SSR
- **About:** Launched browser-based AR trial room with static fallback for low-end devices. Reduced load times **80%**, increased engagement **30–40%**.

Mobile Apps (React Native)

- **Timeline:** 2020 – 2021
- **Tech Stack:** React Native, REST/GraphQL APIs, Multi-Payment Integration
- **About:** Built iOS + Android apps using React Native. Cut dev cycle **30%**, increased reach **40%**.

CRM + Journey Analytics

- **Timeline:** 2019 – 2020
- **Tech Stack:** Django, PostgreSQL, Celery, Grafana, Prometheus, Real-time Session Monitoring
- **About:** Implemented CRM + per-session analytics system. Improved sales efficiency **40%** and reduced lead response times.

E-commerce Core (Ornaz v1)

- **Timeline:** 2018 – 2020
- **Tech Stack:** Django (MVT), PostgreSQL, Redis, RabbitMQ, Celery, HTML, Jinja2, JavaScript, PayU, Razorpay
- **About:** Mobile-first e-commerce platform. Increased mobile sales **40%**, reduced cart drop-offs **25%**, and boosted returning users.